

Machine Learning Assignment

Q1: Create a DataFrame for student marks and print basic information like shape, columns, and data types.

```
data = {  
    'Name': ['Amit', 'Sagar', 'Pooja'],  
    'Math': [85, 90, 78],  
    'Science': [92, 88, 80],  
    'English': [75, 85, 82]  
}
```

Note : Consider the same dataset for this as well as next assignment.

Q2: Use the DataFrame from Q1 and print descriptive statistics using .describe().

Q3: Add a new column 'Total' to the DataFrame as the sum of all subject marks.

Q4: Display students who scored more than 85 in Science.

Q5: Replace 'Pooja' with 'Puja' in the 'Name' column.

Q6: Sort the DataFrame by 'Total' marks in descending order.

Q7: Create a bar plot of student names vs total marks.

Q8: Plot a line chart of marks for 'Amit' across all subjects.

Q9: Create a DataFrame with missing values and fill them with column mean.

```
data2 = {  
    'Name': ['Amit', 'Sagar', 'Pooja'],  
    'Math': [np.nan, 76, 88],  
    'Science': [91, np.nan, 85]  
}
```

Q10: Drop the 'English' column from original DataFrame.

Machine Learning Assignment

- Q1: Normalize the 'Math' scores using Min-Max scaling.
- Q2: Create a gender column and perform one-hot encoding.
- Q3: Group students by gender and calculate average marks.
- Q4: Plot a pie chart of subject marks for 'Sagar'.
- Q5: Add a new column 'Status' where students with total ≥ 250 are 'Pass', else 'Fail'.
- Q6: Count how many students passed.
- Q7: Export the final DataFrame to a CSV file.
- Q8: Plot a histogram of math marks.
- Q9: Rename 'Math' column to 'Mathematics'.
- Q10: Plot a boxplot for English marks to check distribution and outliers.

